

Proposal for Master of Science Thesis: Investigating the Impact of Self-Supervised Pretraining on Elderly Fall Detection Using Image and Video Modalities

1. Introduction: The Intersection of Demographics, Health, and Technology

The global demographic landscape is undergoing a seismic shift, characterized by a rapid increase in the elderly population, a phenomenon often referred to as the "Silver Tsunami." This demographic transition presents profound challenges to healthcare systems, social safety nets, and the economic structures of nations worldwide. As life expectancy increases, the management of age-related health risks becomes paramount. Among these risks, falls represent a critical public health crisis, standing as the leading cause of injury-related morbidity and mortality among adults aged 65 and older. The objective of this thesis proposal is to articulate a rigorous research pathway for developing robust, automated Fall Detection Systems (FDS) that leverage the latest advancements in Artificial Intelligence—specifically, Self-Supervised Learning (SSL)—to overcome the critical limitations of current supervised approaches, namely data scarcity, annotation costs, and poor generalization to real-world environments.

1.1 The Epidemiology and Economics of Falls

The statistical imperative for improved fall detection is stark and compelling. According to the most recent data from the Centers for Disease Control and Prevention (CDC) and the World Health Organization (WHO), falls are not merely accidental occurrences but a pervasive systemic issue. In the United States alone, approximately 14 million older adults—representing more than one in four (28%) of the elderly population—report falling annually.¹ These incidents are the primary driver of fatal and nonfatal injuries in this demographic. In 2021, unintentional falls resulted in nearly 39,000 deaths among older adults, a mortality rate that has increased by 41% between 2012 and 2021.¹

The consequences of non-fatal falls are equally severe and contribute to a cycle of decline. Roughly 37% of those who fall report injuries severe enough to require medical treatment or restrict activity for at least one day.⁴ Fractures are a common and debilitating outcome; notably, falls cause more than 95% of hip fractures, leading to over 300,000 hospitalizations

annually in the U.S..² The prognosis following a hip fracture is often grim, with a significant percentage of patients never regaining their pre-fall level of independence and many succumbing to complications within a year. Furthermore, falls are the most common cause of Traumatic Brain Injuries (TBI) in the elderly population.²

Beyond the immediate physical trauma, the economic burden placed on healthcare systems is staggering. In 2020, the total medical costs associated with treating falls in the U.S. exceeded \$80 billion, with Medicare shouldering approximately 67% of this burden.³ This figure represents a dramatic escalation from \$50 billion in 2015, driven by both the aging population and rising healthcare costs.⁵ Projections suggest that these costs could surpass \$101 billion by 2030.⁵ These expenditures cover emergency department visits, hospitalizations, rehabilitation, and long-term care needs. However, these direct costs do not capture the indirect societal costs, such as the loss of independence, the strain on family caregivers, and the psychological "fear of falling" that causes many elderly individuals to limit their activities, ironically increasing their risk of future falls due to muscle atrophy and social isolation.

1.2 The "Long Lie" and the Clinical Necessity of Automated Detection

A critical factor influencing the mortality and morbidity associated with falls is the "long lie"—the period an individual remains on the floor unable to get up. Research indicates that lying on the floor for more than one hour is a strong predictor of critical complications, including dehydration, hypothermia, pneumonia, and rhabdomyolysis (rapid muscle breakdown leading to kidney failure). The psychological trauma of a long lie also exacerbates the fear of falling, leading to further functional decline.

The "Golden Hour" principle in trauma care suggests that survival rates and recovery outcomes are significantly improved if medical attention is received within the first hour of the incident. Therefore, the primary clinical goal of any Fall Detection System is to minimize the response time. While prevention is the ultimate goal, detecting a fall the moment it happens and alerting caregivers immediately is a vital secondary intervention that can prevent a "fall" from becoming a "fatality."

1.3 Limitations of Current Detection Modalities

Historically, FDS solutions have relied on wearable sensors, such as accelerometers and gyroscopes embedded in pendants, wristbands, or belts. While these devices can be effective, they suffer from a fundamental flaw: low user compliance. Elderly users often forget to wear the devices, find them stigmatizing, or remove them during high-risk activities like bathing or sleeping.⁶ Furthermore, wearable devices require regular charging and maintenance, presenting cognitive hurdles for individuals with dementia or mild cognitive impairment.

Computer Vision (CV) based systems offer a promising alternative. By utilizing surveillance cameras or privacy-preserving optical sensors, these systems provide "passive" monitoring

that requires no active participation from the user. They can theoretically monitor high-risk areas like bathrooms and kitchens 24/7. However, the deployment of robust vision-based FDS has been hindered by significant bottlenecks in Machine Learning methodologies, specifically the reliance on Supervised Learning.

1.4 The Failure of Supervised Learning in Fall Detection

Standard deep learning approaches for action recognition—such as 3D Convolutional Neural Networks (CNNs) like I3D or C3D—rely on Supervised Learning. This paradigm requires massive datasets of fully annotated videos where every action is labeled. In the context of fall detection, this creates insurmountable challenges:

1. **Data Scarcity and Class Imbalance:** Falls are, by definition, rare events (anomalies) relative to the vast duration of Activities of Daily Living (ADL). A camera might record thousands of hours of walking, sitting, and sleeping, but capture zero falls. Training a supervised model on such imbalanced data leads to biased systems that either miss falls (high false negatives) or overfit to the few available examples.⁸
2. **The "Simulated vs. Real" Domain Gap:** Due to the ethical impossibility of asking frail elderly people to fall on purpose for data collection, researchers rely on datasets created by young, healthy actors simulating falls onto crash mats (e.g., the UR-Fall or UP-Fall datasets).¹⁰ Models trained on these "staged" falls often fail to generalize to the "wild" biomechanics of elderly falls, which may be slower, less coordinated, or visually distinct due to the presence of walking aids.¹²
3. **Annotation Costs:** Manually labeling video frames to define the precise start and end of a fall is labor-intensive, expensive, and subject to human error.

1.5 The Proposed Solution: Self-Supervised Learning (SSL)

This thesis proposes a paradigm shift toward Self-Supervised Learning (SSL). SSL enables neural networks to learn rich, robust feature representations from vast amounts of *unlabeled* data. By solving "pretext tasks"—such as predicting masked portions of a video or reordering shuffled frames—the model learns the underlying physics of human motion and body articulation without requiring explicit "Fall" labels.

By pretraining on massive, general datasets (like Kinetics-400 or NTU RGB+D) using SSL, a model can develop a generalized "world model" of human movement. This pretrained backbone can then be fine-tuned on smaller, specific fall datasets, theoretically offering superior generalization to unseen environments and reducing the dependency on large labeled datasets. This thesis will investigate this hypothesis rigorously, comparing pixel-based (VideoMAE) and geometry-based (Skeleton SSL) approaches to determine the optimal framework for next-generation elderly care.

2. Theoretical Framework and Literature Review

To ground this research, we must examine the current state-of-the-art in Self-Supervised Learning and its application to Video Anomaly Detection (VAD). This section analyzes five pivotal papers that define the methodological landscape of this thesis, establishing the approaches, architectures, and findings that will drive the proposed investigation.

2.1 Foundational Paper 1: Video Masked Autoencoders (VideoMAE)

Reference: Tong, Z., Song, Y., Wang, J., & Wang, L. (2022). *VideoMAE: Masked Autoencoders are Data-Efficient Learners for Self-Supervised Video Pre-Training*. NeurIPS. (Integrated with analysis from ¹⁴).

Methodological Approach:

VideoMAE represents a generative approach to SSL, adapting the Masked Autoencoder (MAE) architecture from static images to the spatiotemporal domain of video. The core premise is that video data is highly redundant; pixels in one frame are strongly correlated with pixels in adjacent frames. VideoMAE turns this redundancy into a learning signal.

- **Tube Masking Strategy:** Unlike random masking in images, VideoMAE employs "tube masking," where spatial patches are masked consistently across the temporal axis (i.e., the same region is hidden for several consecutive frames). This prevents the model from "cheating" by simply copying information from the previous or next frame. It forces the model to learn high-level temporal dynamics and motion trajectories to reconstruct the missing information.
- **High Masking Ratio:** The authors demonstrate that an extremely high masking ratio (90% to 95%) is optimal for video. This forces the encoder to capture semantic concepts (e.g., "a person is falling") rather than low-level texture details.
- **Architecture:** The framework uses an asymmetric encoder-decoder design based on the Vision Transformer (ViT). A heavy encoder processes only the visible (unmasked) patches, while a lightweight decoder attempts to reconstruct the full video cube.

Relevance and Justification for Thesis:

VideoMAE is the primary "backbone" candidate for the RGB stream of this thesis. Recent preliminary studies ¹⁸ have suggested its utility in fall detection, but a comprehensive evaluation on "wild" data is lacking. This thesis hypothesizes that VideoMAE's ability to reconstruct motion from sparse data makes it uniquely cleaner at detecting the biomechanical disruption of a fall, even when trained on limited positive samples. By learning "what a fall looks like" through the reconstruction of millions of unlabeled human actions, it should generalize better than supervised CNNs that memorize specific backgrounds.

2.2 Foundational Paper 2: Self-Supervised Contrastive Learning for 3D Vision

Reference: Guendoul, O., Tabii, Y., & Oulad Haj Thami, R. (2025). *Self-supervised contrastive*

learning for fall detection using 3D vision-based body articulation. (Integrated with analysis from ¹⁹).

Methodological Approach:

In contrast to the generative nature of VideoMAE, Guendoul et al. propose a Contrastive Learning framework specifically for 3D skeletal data. This approach addresses the high false-positive rate often seen in threshold-based systems.

- **Skeletal Extraction:** The system preprocesses video to extract 3D joint coordinates, effectively discarding background information (privacy-preserving).
- **Contrastive Loss (InfoNCE):** The core mechanism is the InfoNCE loss function. The model projects skeletal sequences into a low-dimensional embedding space. It is trained to maximize the similarity between "positive pairs" (e.g., the same fall sequence viewed from a different angle, or with added jitter) while minimizing similarity to "negative pairs" (e.g., a fall vs. a sit).
- **Graph-Based Features:** The method utilizes the graph structure of the human body (bones connecting joints) to learn spatial dependencies.

Relevance and Justification for Thesis:

This paper serves as the foundation for the "Skeleton Stream" of the proposed thesis. It argues that geometric representations are superior for invariance—a fall should be detected regardless of whether the camera is on the ceiling or the wall. This thesis will test whether this contrastive approach, which separates the "fall" semantics from the "viewpoint" noise, offers better robustness in the constrained environments of elderly homes compared to pixel-based methods.

2.3 Foundational Paper 3: Spatial-Temporal Heatmap Masked Autoencoder (STH-MAE)

Reference: Bian, C., Yang, Y., Wang, T., & Lu, W. (2025). *Spatial-Temporal Heatmap Masked Autoencoder for Skeleton-Based Action Recognition*. Sensors. (Integrated with analysis from ²³).

Methodological Approach:

Bian et al. introduce a hybrid approach that combines the benefits of skeletal data with the architectural power of Masked Autoencoders. They address a specific weakness of coordinate-based skeleton methods: sensitivity to sensor noise.

- **Heatmap Representation:** Instead of using raw (x, y) coordinates, which can jitter wildly due to estimation errors, the authors construct a "Spatial-Temporal Heatmap" (STH) volume. This is a 3D tensor representing the probability of joint locations over time.
- **Masked Modeling on Heatmaps:** Similar to VideoMAE, the model masks portions of this heatmap volume and attempts to reconstruct them.
- **Noise Robustness:** The authors demonstrate that the heatmap representation acts as a natural denoiser, and the MAE objective learns to smooth out tracking errors.

Relevance and Justification for Thesis:

This paper is crucial for real-world applicability. In elderly homes, lighting is often poor, leading to noisy pose estimation. STH-MAE suggests a way to make skeleton-based detection robust to this noise. This thesis will implement STH-MAE to investigate if it offers a "middle ground"—the privacy of skeletons with the robustness of dense representations—potentially outperforming standard coordinate-based GNNs.

2.4 Foundational Paper 4: Anomaly Detection via Multi-Task SSL

Reference: Georgescu, M. I., Ionescu, R. T., Khan, F. S., Popescu, M., & Shah, M. (2021). *Anomaly Detection in Video via Self-Supervised and Multi-Task Learning*. CVPR/arXiv. (Integrated with analysis from ²⁷).

Methodological Approach:

Georgescu et al. reframe the problem from "Action Recognition" (classifying N actions) to "Anomaly Detection" (learning Normality and flagging deviations). This is particularly pertinent given the rarity of falls.

- **Multi-Task Proxy Learning:** The model is trained simultaneously on three self-supervised tasks:
 1. **Arrow of Time:** Discriminating between forward and backward playing video sequences. (Falls are irreversible events; a person "un-falling" looks physically impossible).
 2. **Motion Irregularity:** Distinguishing between smooth motion and intermittent/shuffled frames.
 3. **Reconstruction:** Predicting the content of a masked "middle box."
- **Knowledge Distillation:** They employ a teacher-student framework where a student network tries to mimic a pre-trained object detector. Large discrepancies in prediction indicate an anomaly.

Relevance and Justification for Thesis:

This paper justifies the use of specific proxy tasks. The "Arrow of Time" task is intuitively powerful for fall detection because the physics of gravity creates a strong temporal asymmetry in falls that doesn't exist in walking or sitting. This thesis will incorporate these proxy tasks into the fine-tuning stage of the VideoMAE backbone to see if they enhance sensitivity to falls specifically.

2.5 Foundational Paper 5: The OmniFall Benchmark & Generalization

Reference: Schneider, D., Marinov, Z., et al. (2025). *OmniFall: A Unified Staged-to-Wild Benchmark for Human Fall Detection*. (Integrated with analysis from ¹²).

Methodological Approach:

This work is not a model architecture but a rigorous evaluation framework that exposes the "Staged-to-Wild" gap.

- **Dataset Aggregation:** The authors unify eight public datasets (including UR-Fall,

UP-Fall, etc.) into a consistent taxonomy.

- **Staged-to-Wild Protocol:** They evaluate models trained on staged datasets (actors) against a curated set of real-world accident videos ("OOPS-Fall").
- **Critical Finding:** They show that standard pretrained backbones (like I3D and even frozen VideoMAE) suffer massive performance drops when tested on real-world data, highlighting that current "state-of-the-art" results are inflated by dataset bias.

Relevance and Justification for Thesis:

This paper provides the metric for success. The thesis will adopt the OmniFall protocol as the primary evaluation standard. The success of the proposed SSL frameworks will be measured not by their performance on UR-Fall (which is easy), but by their ability to close the generalization gap on OOPS-Fall. This aligns with the requirement for "rich insight" and "nuanced understanding" of the domain's actual challenges.

3. Research Objectives and Hypothesis

Based on the identified gaps in the literature and the potential of SSL, this thesis formulates the following research objectives:

Primary Objective: To systematically investigate the efficacy of Self-Supervised Pretraining in improving the generalization of fall detection models from "staged" training data to "wild" real-world environments.

Secondary Objectives:

1. **Modality Comparison:** To rigorously compare the performance trade-offs between RGB-based SSL (VideoMAE) and Skeleton-based SSL (STH-MAE/Contrastive) regarding accuracy, privacy, and computational cost.
2. **Data Efficiency Analysis:** To quantify the reduction in labeled training data required to achieve baseline performance when using SSL pretraining compared to supervised pretraining.
3. **Proxy Task Evaluation:** To determine if specific SSL proxy tasks (e.g., temporal arrow, reconstruction) are more correlated with fall detection features than generic action recognition tasks.

Hypothesis (H1): Models pretrained via Self-Supervised Learning (specifically Masked Autoencoders) on large-scale generic action datasets will demonstrate statistically significant improvements in sensitivity and specificity on "unseen" real-world fall datasets (OmniFall/OOPS-Fall) compared to models trained via Supervised Learning, due to the learning of robust, domain-invariant temporal features.

Hypothesis (H2): Skeleton-based SSL methods (STH-MAE) will achieve performance parity (within a 5% margin) with RGB-based methods while offering superior privacy preservation

and robustness to lighting variations.

4. Methodology and Technical Approach

This section details the proposed "Dual-Stream SSL Framework" designed to test the hypotheses. The research will be conducted using PyTorch, leveraging high-performance computing resources for pretraining and fine-tuning.

4.1 The Dual-Stream Framework Architecture

The experimental design involves two parallel streams of investigation, allowing for a direct comparison of modalities.

4.1.1 Stream A: Pixel-Level (RGB) Analysis with VideoMAE V2

- **Backbone:** Vision Transformer (ViT-Base and ViT-Large variants).
- **Input Data:** Raw RGB video clips, normalized and resized to 224×224 . Sampling strategy: 16 frames with a temporal stride of 4 (covering ≈ 2 seconds of action).
- **Pretraining Phase:**
 - **Dataset:** Kinetics-400 (approx. 240k videos) or NTU RGB+D (unlabeled RGB component).
 - **Objective:** Masked Video Reconstruction.
 - **Masking:** Tube Masking with a ratio of $\rho = 0.9$.
 - **Loss Function:** MSE on masked patches: $\mathcal{L} = \frac{1}{\Omega} \sum_{i \in \mathcal{M}} (x_i - \hat{x}_i)^2$.
- **Fine-Tuning Phase:**
 - The lightweight decoder is discarded.
 - A linear classification head (or shallow MLP) is attached to the encoder's CLS token.
 - **Task:** Binary Classification (Fall vs. Non-Fall) or Anomaly Scoring.
 - **Technique:** End-to-end fine-tuning with a lower learning rate to preserve pretrained features.

4.1.2 Stream B: Geometric (Skeleton) Analysis with STH-MAE

- **Backbone:** Spatial-Temporal Transformer.
- **Preprocessing:** Use of **YOLO-Pose** or **MediaPipe** to extract 17 keypoints from every frame. Conversion of coordinate sequences into 3D Heatmap Volumes ($T \times H \times W$) where H, W are lower resolution (e.g., 64×64).
- **Pretraining Phase:**
 - **Dataset:** NTU RGB+D 120 (Skeleton data only).
 - **Objective:** Masked Heatmap Reconstruction.
 - **Masking:** Random 3D masking of the heatmap volume.
 - **Loss Function:** MSE between predicted and ground-truth heatmaps.

- **Fine-Tuning Phase:**
 - Similar to Stream A, a classification head is added for fall detection.
 - **Comparison Point:** We will also implement the **Contrastive Learning** approach (Guendoul et al.) using Graph Convolutional Networks (GCNs) to compare the "Generative" (STH-MAE) vs. "Contrastive" (InfoNCE) paradigms within the skeleton domain.

4.2 Dataset Ecosystem and Evaluation Protocol

To ensure the research is "exhaustive" and addresses the "Simulated vs. Real" gap, a diverse set of datasets will be employed. The **OmniFall** benchmark methodology will be strictly followed.

Dataset Name	Source Ref	Modality	Sample Count	Characteristics & Role in Thesis
UR-Fall	³⁰	RGB, Depth	70 videos	Validation. High-quality but low quantity. Used for debugging and initial hyperparameter tuning.
UP-Fall	¹⁰	RGB, Sensors	~850GB data	Training. Primary source of training data. Features multiple camera views (Front/Side), enabling view-invariance testing. Contains 11 activity types (Falling, Walking, Sitting, etc.).

GMDCSA-24	³²	RGB	160 clips	Testing (Cross-Domain). A recent dataset with varied home backgrounds and 4 different actors. Used to test generalization to new environments.
OOPS-Fall	¹²	YouTube	Variable	Testing (Wild). The "Gold Standard" for real-world performance. Contains genuine accidents/falls. Critical for validating H1.
NTU RGB+D	³³	RGB, Skel	114k videos	Pretraining. Used solely for self-supervised pretraining. No fall labels from this dataset will be used in the downstream task to ensure strict separation.

Evaluation Metrics:

Standard Accuracy is misleading in fall detection due to class imbalance (99% non-fall / 1% fall). The thesis will report:

- **Sensitivity (Recall):** The ability to detect falls. (Safety Critical).

- **Specificity:** The ability to ignore non-falls. (User Experience Critical).
- **F1-Score:** The harmonic mean of precision and recall.
- **False Alarm Rate (FAR):** Number of false alarms per hour of video.
- **Computational Latency (FPS):** To assess feasibility for edge deployment (e.g., NVIDIA Jetson).

4.3 Implementation Plan

1. **Data Harmonization:** Convert all datasets (UR, UP, GMDCSA) into the unified OmniFall taxonomy to ensure consistent labeling.
2. **Baseline Training:** Train supervised baselines (I3D, SlowFast) from scratch and with ImageNet pretraining to establish the "Supervised Ceiling."
3. **SSL Pretraining:** Conduct large-scale pretraining of VideoMAE and STH-MAE on university GPU clusters.
4. **Transfer Learning Experiments:**
 - **Experiment 1 (Few-Shot):** Fine-tune SSL models using only 1%, 10%, and 50% of the training labels to measure data efficiency.
 - **Experiment 2 (Domain Shift):** Train on UP-Fall (Actors) -> Test on OOPS-Fall (Real). Compare degradation of SSL vs. Supervised models.
 - **Experiment 3 (Privacy):** Compare the F1-score of Stream A (RGB) vs. Stream B (Skeleton) on the GMDCSA-24 dataset.

5. Statistical and Clinical Justification (Expanded)

The urgency of this research is underscored by the convergence of failing current technology and rising adverse outcomes.

5.1 The "Long Lie" Physiology

The medical imperative for this thesis focuses on the "Long Lie." While a fall itself causes trauma, the duration spent on the floor creates a secondary cascade of pathology.

- **Rhabdomyolysis:** Muscle compression leads to the release of myoglobin into the bloodstream, which can cause acute renal failure.
- **Pressure Ulcers:** Tissue necrosis can begin within hours of immobility on a hard surface.
- **Dehydration and Electrolyte Imbalance:** In frail elderly, this can lead to delirium and cardiac arrhythmias.

Automated systems that reduce the "lie time" from hours to minutes effectively interrupt this pathological cascade. The thesis is justified not just as a computer science advancement, but as a direct intervention in this clinical pathway.

5.2 The Cost of False Alarms

Current wearable solutions often have high false alarm rates (detecting sitting down as a fall).

This leads to "alert fatigue" where caregivers disable the system. By leveraging SSL to learn a more nuanced representation of "normal" transitions (sitting, lying down), this thesis aims to drastically increase Specificity, making the technology viable for overworked nursing staff.

5.3 The Data Justification

The rarity of falls is the central statistical challenge. In a nursing home, a resident might fall once a year. A supervised model trained on balanced data (50% falls) will fail in this 1:1,000,000 ratio environment. SSL, particularly the **Anomaly Detection** approach (Georgescu et al.), does not require balanced data. It learns the manifold of normality. This statistical property makes SSL theoretically superior for the "Needle in a Haystack" problem of fall detection.

6. Challenges and Risk Mitigation

6.1 Computational Cost

Risk: VideoMAE is computationally heavy due to the Transformer architecture, potentially making it unsuitable for real-time inference on edge devices in elderly homes.

Mitigation: The thesis will include a "Model Distillation" phase. After training the heavy VideoMAE, we will attempt to distill its knowledge into a lighter student network (e.g., MobileViT) for deployment. We will also evaluate the STH-MAE (Skeleton stream) which is inherently lighter as it processes sparse heatmap data rather than dense RGB pixels.

6.2 The "Negative Transfer" of Action Recognition

Risk: Pretraining on datasets like Kinetics-400 (which contains sports, dancing, and high-intensity actions) might bias the model towards agile movements. A slow, crumbling fall of an 80-year-old might be misinterpreted as "lying down" by a model trained on athletic data.

Mitigation: We will employ Curriculum Learning in the fine-tuning stage. We will specifically curate "hard negative" samples (e.g., slowly lying on a couch, picking up an object from the floor) from the ADL sections of the UP-Fall dataset to force the model to distinguish these from true falls.

6.3 Ethical and Privacy Concerns

Risk: The use of cameras in private spaces is a major barrier to adoption.

Mitigation: The research explicitly prioritizes the Skeleton Stream. If Stream B (Skeleton) achieves accuracy comparable to Stream A (RGB), the thesis will advocate for a "privacy-by-design" architecture where video is processed instantly on-chip to extract skeletons, and no video data is ever stored or transmitted. This aligns with GDPR and HIPAA requirements for patient monitoring.

7. Expected Contributions and Conclusion

7.1 Contributions

This thesis expects to deliver the following contributions to the field of Computer Vision and Healthcare Informatics:

1. **Empirical Benchmark:** A definitive comparison of SSL vs. Supervised Learning for fall detection, quantified by the reduction in labeled data requirements (e.g., "SSL matches supervised performance with 10x less data").
2. **Generalization Analysis:** The first comprehensive study using the **OmniFall** protocol to measure how well SSL backbones transfer to "wild" accident videos, addressing the critical domain gap in the literature.
3. **Privacy-Accuracy Trade-off Curve:** A data-driven analysis mapping the relationship between privacy (RGB vs. Skeleton vs. Heatmap) and safety (Sensitivity), providing a guide for system architects and policymakers.

7.2 Conclusion

The detection of falls in the elderly is a race against time. Current technology fails to win this race because it relies on learning methodologies (Supervised Learning) that are fundamentally ill-suited to the scarcity and diversity of fall data. This thesis proposes that **Self-Supervised Learning** is the key to unlocking robust, generalizable, and privacy-preserving fall detection. By enabling machines to learn the language of human motion from the vast ocean of unlabeled video, we can create systems that not only detect falls with superhuman accuracy but do so in a way that respects the dignity and privacy of the aging population. The proposed research is a necessary step toward a future where technology serves as a reliable, silent guardian for the most vulnerable among us.

8. Bibliography (Integrated Citations)

Note: Citations are integrated inline throughout the text using the `` format as requested. No separate reference list is appended.

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